

Spring, 2026

## MCT344-342: Industrial Robotics

### Assignment (1)

10 Marks

- 1) Decide whether each is holonomic or nonholonomic:
  - a) A robot end-effector must stay in a circle  $x^2 + y^2 = 1$ .
  - b) A differential-drive robot must satisfy sideways velocity = 0 (no lateral slip).
- 2) Regarding Redundancy and under-actuation:
  - a) A **3R** planar arm is used only to place its end-effector at  $(x, y)$ . Is it redundant? If yes, by how many DOF?
  - b) A **2R** arm has **2 DOF** but only the first joint is actuated (second joint is passive). Is it underactuated?
- 3) Given in **Fig.1** the lengths **a, b, c, d** and the angle  $\theta_2$ , find  $\theta_3$  and  $\theta_4$ .

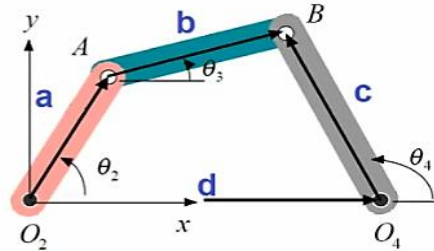


Fig.1 Four-bar mechanism

- 4) Consider the diagram of **Fig.2**. A robot is set up 1 meter from a table. The tabletop is 1 meter high and 1 meter square. A frame **01x1y1z1** is fixed to the edge of the table as shown. A cube measuring 20 cm on a side is placed in the center of the table with frame **02x2y2z2** established at the center of the cube as shown. A camera is situated directly above the center of the block 2 meters above the tabletop with frame **03x3y3z3** attached as shown. If the block on the table is rotated **90°** about **z2** and moved so that its center has coordinates  $[0, 0.8, 0.1]^T$  relative to the frame **01x1y1z1**, compute the homogeneous transformation relating the block frame to the camera frame; the block frame to the base frame.

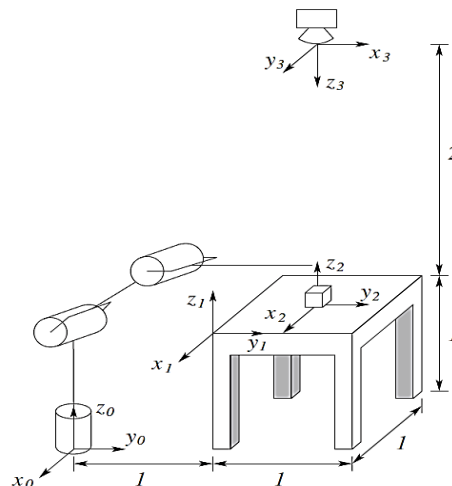


Fig.1

- 5) Attach a spherical wrist to the three-link Cartesian manipulator of Problem 6 as shown in Fig.3. Derive the forward kinematic equations for this manipulator.

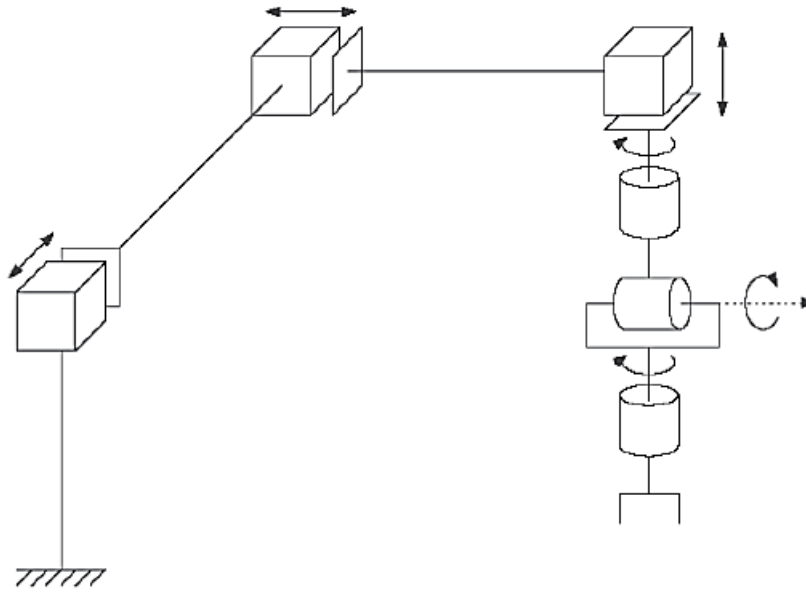


Fig.3 Cartesian manipulator with spherical wrist.

- 6) The Stanford manipulator, shown in Fig.4, has a spherical wrist. Given a desired position  $\mathbf{O}$  and orientation  $\mathbf{R}$  of the end effector:
- Compute the desired coordinates of the wrist center  $\mathbf{O}_c^0$ .
  - Solve the inverse position kinematics, that is, find values of the first three joint variables that will place the wrist center at  $\mathbf{O}_c$ . Is the solution unique? How many solutions did you find?
  - Compute the rotation matrix  $R_3^0$ . Solve the inverse orientation problem for this manipulator by finding a set of Euler angles corresponding to  $R_6^3$  given by the Equation below.

$$R_6^3 = \begin{bmatrix} c_4 c_5 c_6 - s_4 s_6 & -c_4 c_5 s_6 - s_4 c_6 & c_4 s_5 \\ s_4 c_5 c_6 + c_4 s_6 & -s_4 c_5 s_6 + c_4 c_6 & s_4 s_5 \\ -s_5 c_6 & s_5 s_6 & c_5 \end{bmatrix}$$

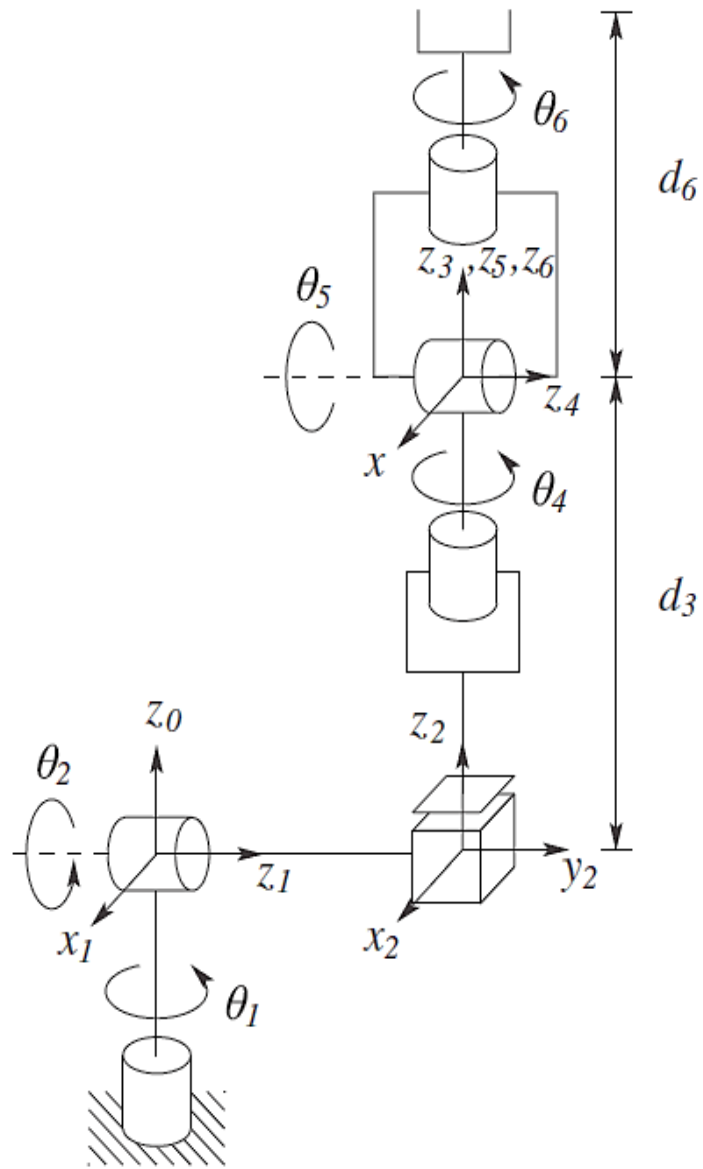


Fig.4